MA1522 Linear Algebra for Computing

Lesson 15: QR Factorization and Least Squares Solutions

Announcements

- No class on Friday 9 October (Well-Being Day), nor on Tuesday 13 and Friday 16 October. The next lesson is **Tuesday 20 October**.

- **Online Quiz 4** opens **Saturday 17 October, 0:00**, and is due **Sunday 18 October, 23:59**.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson15` (e.g. `A1234567Z_lesson15.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 15**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

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Question 1

$$A = \begin{pmatrix} 1 & -3 & -1 \\ 0 & 2 & 4 \\ 2 & -2 & -3 \\ 2 & -1 & -1 \end{pmatrix} = (\,\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3\,).$$

The Gram-Schmidt process applied to $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ gives the orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for $\mathrm{Col}(A)$, where

$$\mathbf{v}_1 = \mathbf{a}_1, \quad \mathbf{v}_2 = \mathbf{a}_2 + \mathbf{v}_1, \quad \mathbf{v}_3 = \mathbf{a}_3 + \mathbf{v}_1 - \mathbf{v}_2.$$

1. Find two upper triangular matrices R_1, R_2 such that

$$(\,\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3\,)\,R_1 = (\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{a}_3\,), \qquad (\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{a}_3\,)\,R_2 = (\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3\,).$$

2. Deduce that $A = QR$, where Q is orthogonal and R is upper triangular with positive diagonal entries.

Question 1 — solution: part 1

1. Column j of each R_i records how the j-th new column is made from the old columns. Since $\mathbf{v}_1 = \mathbf{a}_1$, $\mathbf{v}_2 = \mathbf{a}_1 + \mathbf{a}_2$, so

$$(\,\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3\,)\begin{pmatrix}1 & 1 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix} = (\,\mathbf{a}_1\ \ \mathbf{a}_1+\mathbf{a}_2\ \ \mathbf{a}_3\,) = (\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{a}_3\,), \qquad R_1 = \begin{pmatrix}1 & 1 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}.$$

Next, $\mathbf{v}_3 = \mathbf{v}_1 - \mathbf{v}_2 + \mathbf{a}_3$, so

$$(\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{a}_3\,)\begin{pmatrix}1 & 0 & 1\\ 0 & 1 & -1\\ 0 & 0 & 1\end{pmatrix} = (\,\mathbf{v}_1\ \ \mathbf{v}_2\ \ \mathbf{v}_1-\mathbf{v}_2+\mathbf{a}_3\,) = (\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3\,), \qquad R_2 = \begin{pmatrix}1 & 0 & 1\\ 0 & 1 & -1\\ 0 & 0 & 1\end{pmatrix}.$$

Hence $AR_1R_2 = (\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3\,)$: Gram-Schmidt is multiplication by upper triangular matrices.

Question 1 — solution: part 2

2. Deduce that $A = QR$, where Q is orthogonal and R is upper triangular with positive diagonal entries.

Normalise the orthogonal basis from part 1 (each $\mathbf{v}_i \neq \mathbf{0}$). Since $(\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3\,) = AR_1R_2$,

$$Q = \begin{pmatrix} \dfrac{\mathbf{v}_1}{\|\mathbf{v}_1\|} & \dfrac{\mathbf{v}_2}{\|\mathbf{v}_2\|} & \dfrac{\mathbf{v}_3}{\|\mathbf{v}_3\|} \end{pmatrix} = A\,R_1R_2\,D, \qquad D = \begin{pmatrix} \|\mathbf{v}_1\|^{-1} & 0 & 0 \\ 0 & \|\mathbf{v}_2\|^{-1} & 0 \\ 0 & 0 & \|\mathbf{v}_3\|^{-1} \end{pmatrix}.$$

Its columns are orthogonal unit vectors, so $Q^TQ = I_3$. Hence $A = QR$ with

$$R = (R_1R_2D)^{-1} = D^{-1}R_2^{-1}R_1^{-1} = D^{-1}\begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \|\mathbf{v}_1\| & -\|\mathbf{v}_1\| & -\|\mathbf{v}_1\| \\ 0 & \|\mathbf{v}_2\| & \|\mathbf{v}_2\| \\ 0 & 0 & \|\mathbf{v}_3\| \end{pmatrix}.$$

R is upper triangular with positive diagonal entries. ✓ □

Section 5.4 QR Factorization

Theorem (QR Factorization)

Suppose A is an $m \times n$ matrix with linearly independent columns. Then A can be written as

$$A = QR$$

for some $m \times n$ matrix Q such that $Q^T Q = I_n$ and invertible upper triangular matrix R with positive diagonal entries. This decomposition is called a **QR factorization** of A.

Algorithm to QR Factorization

Let $A = (\,\mathbf{a}_1\ \mathbf{a}_2\ \cdots\ \mathbf{a}_n\,)$ have linearly independent columns.

1. Perform Gram-Schmidt on $\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$ to obtain an orthonormal set $\{\mathbf{q}_1, \ldots, \mathbf{q}_n\}$.
2. Set $Q = (\,\mathbf{q}_1\ \mathbf{q}_2\ \cdots\ \mathbf{q}_n\,)$.
3. Compute $R = Q^T A$.

Exercise: Write Q and R explicitly for Question 1.

Challenge 1

Let A be an $m \times n$ matrix with linearly independent columns. Show that the QR factorization of A is unique: if $A = QR = Q'R'$, where $Q^T Q = Q'^T Q' = I_n$ and R, R' are upper triangular with positive diagonal entries, then $Q = Q'$ and $R = R'$.

Section 5.5 Least Squares Approximation

Let A be an $m \times n$ matrix and $\mathbf{b}$ a vector in $\mathbb{R}^m$.

Definition

A vector $\mathbf{u}$ in $\mathbb{R}^n$ is a **least squares solution** of $A\mathbf{x} = \mathbf{b}$ if for every vector $\mathbf{v} \in \mathbb{R}^n$,

$$\|A\mathbf{u} - \mathbf{b}\| \leq \|A\mathbf{v} - \mathbf{b}\|.$$

Theorem

A vector $\mathbf{u}$ in $\mathbb{R}^n$ is a least squares solution to $A\mathbf{x} = \mathbf{b}$ if and only if $A\mathbf{u}$ is the projection of $\mathbf{b}$ onto $\mathrm{Col}(A)$.

Section 5.5 Least Squares Solutions and $A^TA\mathbf{x} = A^T\mathbf{b}$

Let A be an $m \times n$ matrix and $\mathbf{b}$ a vector in $\mathbb{R}^m$.

Theorem

A vector $\mathbf{u}$ in $\mathbb{R}^n$ is a least squares solution to $A\mathbf{x} = \mathbf{b}$ if and only if $\mathbf{u}$ is a solution to $A^TA\mathbf{x} = A^T\mathbf{b}$.

Proof.

$$
\begin{aligned}
\mathbf{u} \text{ is a least squares solution to } A\mathbf{x} = \mathbf{b} &\iff A\mathbf{u} \text{ is the orthogonal projection of } \mathbf{b} \text{ onto } \mathrm{Col}(A) \\
&\iff A\mathbf{u} - \mathbf{b} \text{ is orthogonal to } \mathrm{Col}(A) \\
&\iff A^T(A\mathbf{u} - \mathbf{b}) = \mathbf{0} \\
&\iff A^TA\mathbf{u} = A^T\mathbf{b}. \qquad \square
\end{aligned}
$$

Question 2

Let

$$B = \begin{pmatrix} -1 & 1 & 0 \\ 2 & 1 & 3 \\ -2 & 2 & 0 \\ -1 & 2 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} -4 \\ 1 \\ -3 \\ -3 \end{pmatrix}.$$

1. Find all least squares solutions of $B\mathbf{x} = \mathbf{b}$.

2. Find the projection of $\mathbf{b}$ onto $\mathrm{Col}(B)$. Is $B\mathbf{x} = \mathbf{b}$ consistent?

Question 2 — solution: part 1

1. Solve $B^TB\mathbf{x} = B^T\mathbf{b}$. The entries are inner products of the columns of B with each other and with $\mathbf{b}$:

$$(B^TB \mid B^T\mathbf{b}) = \left(\begin{array}{ccc|c} 10 & -5 & 5 & 15 \\ -5 & 10 & 5 & -15 \\ 5 & 5 & 10 & 0 \end{array}\right) \xrightarrow{\text{rref}} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

With $x_3 = t$, the least squares solutions are

$$\mathbf{u} = \begin{pmatrix} 1-t \\ -1-t \\ t \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}, \qquad t \in \mathbb{R}.$$

Question 2 — solution: part 2

2. By the theorem, $B\mathbf{u}$ is the projection of $\mathbf{b}$ onto $\mathrm{Col}(B)$, for any least squares solution $\mathbf{u}$. Since $B(-1,-1,1)^T = \mathbf{0}$,

$$B\mathbf{u} = B\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -2 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ -4 \\ -3 \end{pmatrix} \quad \text{for every } t.$$

$\mathbf{b} - B\mathbf{u} = (-2,0,1,0)^T \neq \mathbf{0}$, so $\mathbf{b} \notin \mathrm{Col}(B)$ and $B\mathbf{x} = \mathbf{b}$ is inconsistent.

$(-2,0,1,0)^T$ is orthogonal to every column of B. ✓

The least squares solutions are not unique, but the projection is. □

Section 5.5 Least Squares Solutions by QR Factorization

Theorem

Let A be an $m \times n$ matrix with linearly independent columns, with QR factorization $A = QR$, and let $\mathbf{b} \in \mathbb{R}^m$. Then $A\mathbf{x} = \mathbf{b}$ has a unique least squares solution,

$$\mathbf{u} = R^{-1}Q^T\mathbf{b}.$$

That is, it suffices to solve

$$R\mathbf{x} = Q^T\mathbf{b}.$$

Proof.

$$\begin{aligned}
\mathbf{u} \text{ is a least squares solution to } A\mathbf{x} = \mathbf{b} &\iff A^TA\mathbf{u} = A^T\mathbf{b} \\
&\iff R^T(Q^TQ)R\mathbf{u} = R^TQ^T\mathbf{b} \\
&\iff R\mathbf{u} = Q^T\mathbf{b} \\
&\iff \mathbf{u} = R^{-1}Q^T\mathbf{b}. \qquad \square
\end{aligned}$$

Question 3

Let A be the matrix of Question 1. It has the QR factorization $A = QR$, where

$$Q = \frac{1}{3}\begin{pmatrix} 1 & -2 & 2 \\ 0 & 2 & 2 \\ 2 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 3 & -3 & -3 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{pmatrix}.$$

Let $\mathbf{b} = \begin{pmatrix} 3 \\ -1 \\ -5 \\ -1 \end{pmatrix}$. Use the QR factorization to find the least squares solution $\mathbf{u}$ of $A\mathbf{x} = \mathbf{b}$.

Question 3 — solution

By the theorem on slide 15, solve $R\mathbf{x} = Q^T\mathbf{b}$. First,

$$Q^T\mathbf{b} = \frac{1}{3}\begin{pmatrix} 1 & 0 & 2 & 2 \\ -2 & 2 & 0 & 1 \\ 2 & 2 & -1 & 0 \end{pmatrix}\begin{pmatrix} 3 \\ -1 \\ -5 \\ -1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 3+0-10-2 \\ -6-2+0-1 \\ 6-2+5+0 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \\ 3 \end{pmatrix}.$$

Back substitution, last unknown first:

$$\begin{pmatrix} 3 & -3 & -3 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \\ 3 \end{pmatrix}: \qquad \begin{aligned} 3x_3 &= 3, & x_3 &= 1, \\ 3x_2 + 3 &= -3, & x_2 &= -2, \\ 3x_1 + 6 - 3 &= -3, & x_1 &= -2. \end{aligned}$$

So $\mathbf{u} = (-2, -2, 1)^T$. □

Next lesson

Next lesson (**Tue 20 Oct**, Section 6.1), I will focus on

1. the definition of eigenvalues and eigenvectors of a square matrix, and how to find them using the characteristic polynomial;

2. the **eigenspace** associated to an eigenvalue, and how to find a basis for it.

Please watch the video clips beforehand.

Challenge 1 — solution

Put $T = R'R^{-1}$. Products and inverses of upper triangular matrices are upper triangular, so T is upper triangular, with positive diagonal entries r'_{ii}/r_{ii}. From $QR = Q'R'$ we get $Q = Q'R'R^{-1} = Q'T$, so

$$I_n = Q^T Q = T^T Q'^T Q' T = T^T T.$$

Hence $T^T = T^{-1}$. The left side is lower triangular and the right side is upper triangular, so T is diagonal, and then $T^T T = I_n$ gives $t_{ii}^2 = 1$. Since $t_{ii} > 0$, $T = I_n$: that is, $R' = R$ and $Q = Q'$.

Without "positive", uniqueness fails: $A = QR = (-Q)(-R)$. □